PYTHON WITH AI • PROJECT REPORT

Student Performance Analytics System

Python Project Documentation

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Course	Python with AI
Institute / Training Centre	EWB Courses
Technologies	Python, Pandas, NumPy
GitHub Repository	https://github.com/ansh-codes-java/Student-Performance-Analytics-System.git

1. Project Objective

For this project I wrote a Python program that takes student marks and attendance and turns them into a report that is easy to read. The report shows how each student did, how the subjects and departments compare, who the top performers are, and which students might need some extra support with their studies.

2. Technologies Used

Technology	What I used it for
Python	The main program logic: functions, loops and if/else conditions.
Pandas	Keeping the records in a DataFrame and doing the grouping and summaries.
NumPy	Numerical calculations where they were needed.
CSV files	Storing the student data and saving the final report.

3. Dataset Description

The dataset has 20 student records and 9 columns: Student_ID, Name, Department, Physics, Maths, Chemistry, Communication, Programming and Attendance. I checked it for missing values and found none. The five subject marks are used to work out each student's total, average and grade, and the attendance column is used to flag students who fall below 75%.

4. How the Program Works

1. **Load and check the data.** The program prints the number of records and columns, the column names, the count of missing values and the first five rows.
2. **Build the student-wise report.** For every student it shows the total, average, percentage, grade, pass/fail result and attendance.
3. **Summarize the class.** It gives the number of students who passed and failed, the pass percentage, the class average, median, standard deviation and the average attendance.
4. **Look at each subject.** It compares subject averages, highest and lowest marks, the topper in each subject and how many students passed or failed.
5. **Rank the students.** The top five students are listed according to their average marks.
6. **Compare departments and grades.** It shows each department's student count, average marks, average attendance and pass count, plus the overall grade distribution.
7. **Find students who need attention.** It lists the students who failed a subject and those whose attendance is below 75%.
8. **Save the report.** Finally, the report is written to results.csv.

5. Key Features

- Dataset overview with a missing-value check
- Student-wise total, average, percentage, grade and pass/fail result
- Class-level statistical summary
- Subject-wise performance comparison
- Top five performers
- Grade distribution
- Department-wise analysis
- List of failed students and students with attendance below 75%
- Report exported to results.csv

6. Results at a Glance

These are the main numbers from the program's output:

Metric	Result
Total students	20
Dataset columns	9
Missing values	0
Passed / Failed	17 / 3
Pass percentage	85.0%
Class average	68.98
Class median	70.3
Standard deviation	15.79
Average attendance	83.5%
Highest average	Pooja Tiwari - 93.0
Lowest average	Vikram Yadav - 42.0
Best subject (by average)	Communication - 71.75
Weakest subject (by average)	Maths - 66.85

7. Output Screenshots

The screenshots below are from the actual output of my program, in the order the report prints them.

```
1. DATASET OVERVIEW
============================================================
Number of students : 20
Number of columns  : 9
Column names       : ['Student_ID', 'Name', 'Department', 'Physics', 'Maths', 'Chemistry', 'Communication', 'Programming', 'Attendan
ce']
Missing values     : 0

First 5 records:
   Student_ID          Name Department  Physics  Maths  Chemistry  Communication  Programming  Attendance
0           1  Aarav Sharma        CSE       85     78         82             74           91          92
1           2    Riya Verma        CSE       45     38         52             60           48          70
2           3   Kabir Singh        ECE       67     72         58             65           70          85
3           4  Ananya Gupta        CSE       92     88         90             85           95          97
4           5   Rohan Patel         ME       55     48         62             58           50          74

============================================================
2. STUDENT-WISE REPORT
============================================================
 Student_ID          Name Department  Total  Average  Percentage Grade Result  Attendance Attendance_Status
          1  Aarav Sharma        CSE    410     82.0        82.0     A   PASS          92                OK
          2    Riya Verma        CSE    243     48.6        48.6     F   FAIL          70         SHORTAGE
          3   Kabir Singh        ECE    332     66.4        66.4     C   PASS          85                OK
          4  Ananya Gupta        CSE    451     90.2        90.2    A+   PASS          97                OK
          5   Rohan Patel         ME    273     54.6        54.6     D   PASS          74         SHORTAGE
          6   Sneha Joshi        ECE    386     77.2        77.2     B   PASS          89                OK
          7  Vikram Yadav         ME    210     42.0        42.0     F   FAIL          60         SHORTAGE
          8  Priya Mishra        CSE    434     86.8        86.8     A   PASS          95                OK
          9   Arjun Reddy         EE    315     63.0        63.0     C   PASS          81                OK
         10  Neha Agarwal         EE    359     71.8        71.8     B   PASS          88                OK
```

Figure 1. Dataset overview and the start of the student-wise report.

```
          7  Vikram Yadav         ME    210     42.0        42.0     F   FAIL          60         SHORTAGE
          8  Priya Mishra        CSE    434     86.8        86.8     A   PASS          95                OK
          9   Arjun Reddy         EE    315     63.0        63.0     C   PASS          81                OK
         10  Neha Agarwal         EE    359     71.8        71.8     B   PASS          88                OK
         11 Karan Malhotra        CE    262     52.4        52.4     D   PASS          72         SHORTAGE
         12   Pooja Tiwari       CSE    465     93.0        93.0    A+   PASS          98                OK
         13 Aditya Chauhan        CE    300     60.0        60.0     C   PASS          79                OK
         14    Simran Kaur       ECE    405     81.0        81.0     A   PASS          91                OK
         15   Rahul Dubey         ME    213     42.6        42.6     F   FAIL          65         SHORTAGE
         16    Isha Saxena        EE    359     71.8        71.8     B   PASS          86                OK
         17   Manish Kumar        CE    344     68.8        68.8     C   PASS          83                OK
         18   Divya Pandey       CSE    420     84.0        84.0     A   PASS          93                OK
         19     Harsh Jain       ECE    275     55.0        55.0     D   PASS          76                OK
         20  Tanvi Rathore         EE    442     88.4        88.4     A   PASS          96                OK

============================================================
3. CLASS SUMMARY
============================================================
Total students     : 20
Passed             : 17
Failed             : 3
Pass percentage    : 85.0 %
Class average      : 68.88
Class median       : 70.3
Standard deviation : 15.79
Highest average    : Pooja Tiwari - 93.0
Lowest average     : Vikram Yadav - 42.0
Average attendance : 83.5 %
```

Figure 2. Rest of the student-wise results and the class summary.

```
============================================================
4. SUBJECT-WISE ANALYSIS
============================================================
      Subject  Average  Highest  Lowest        Topper  Passed  Failed
      Physics    68.55       95      35  Pooja Tiwari      19       1
        Maths    66.85       91      36  Pooja Tiwari      18       2
    Chemistry    67.95       93      40  Pooja Tiwari      20       0
Communication    71.75       92      50 Tanvi Rathore      20       0
  Programming    69.80       98      38  Pooja Tiwari      19       1

Best subject    : Communication (average 71.75 )
Weakest subject : Maths (average 66.85 )

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5. TOP 5 PERFORMERS
============================================================
   Student_ID           Name Department  Total  Average Grade
1          12   Pooja Tiwari        CSE    465     93.0    A+
2           4   Ananya Gupta        CSE    451     90.2    A+
3          20  Tanvi Rathore         EE    442     88.4     A
4           8   Priya Mishra        CSE    434     86.8     A
5          18   Divya Pandey        CSE    420     84.0     A

============================================================
6. GRADE DISTRIBUTION
============================================================
A+ : 2
A : 5
B : 3
C : 4
```

Figure 3. Subject-wise analysis, top five performers and grade distribution.

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7. DEPARTMENT-WISE ANALYSIS
============================================================
            Students  Avg_Marks  Avg_Attendance  Passed
Department
CSE                6      80.77           90.83       5
EE                 4      73.75           87.75       4
ECE                4      69.90           85.25       4
CE                 3      60.40           78.00       3
ME                 3      46.40           66.33       1

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8. STUDENTS NEEDING ATTENTION
============================================================
Failed students:
  - Riya Verma failed in: Maths
  - Vikram Yadav failed in: Physics Programming
  - Rahul Dubey failed in: Maths

Students with attendance below 75 %:
  - Riya Verma : 70 %
  - Rohan Patel : 74 %
  - Vikram Yadav : 60 %
  - Karan Malhotra : 72 %
  - Rahul Dubey : 65 %

Full report saved in results.csv
```

Figure 4. Department-wise analysis, students needing attention and the CSV export message.

8. Final Outcome

The program takes 20 raw student records and turns them into a clear performance report. It brings marks and attendance together, shows who passed and who failed, compares subjects and departments, ranks the top performers and points out the students who may need extra help. The whole report is also saved as a CSV file so it can be used again later.

9. Challenges and Learning

This project gave me hands-on practice with organizing data in tables, going through many student records at once, using functions and conditions, calculating performance measures, grouping data for analysis and presenting the results in a clean, readable way.

10. GitHub Repository

The full source code for this project is on GitHub:

Repository URL: https://github.com/ansh-codes-java/Student-Performance-Analytics-System.git

11. Conclusion

The Student Performance Analytics System shows how Python, Pandas and NumPy can be used to analyze a student dataset and produce useful academic summaries. It presents both individual and overall performance in an organized way, and it makes it easy to spot the strong performers as well as the students who may need support.